Research on Drone Forensics

Ramiz Aliguliyev
COM6016: Cyber Threat Hunting And Digital Forensics
The University of Sheffield

November 2025

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1 Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, modern technology in aircraft, without the need for a pilot, can be operated remotely or by autonomous flight.to the target in military or civilian needs [1]. UAVs can be used as weapons in military for explosion of main military facilities or can be used in modern society for civilian fields. UAVs significantly enhance the monitoring and observation of constructions and sites, helps in project management and safety protocols [1]. Additionally, military-grade UAVs are built for accuracy and strategic applications and play a major role in national security [1]. However, UAVs also can be used in criminal attributes, for example if done deliver drugs to the prison or flies around the airport, the police have only physical machine [1]. Forensics is only way to link that machine to a specific sender [1]. It can reveal the start point and the path of the drone, and often the pilot's email address if he was synchronized with the cloud [1]. Therefore drone forensics is a critical discipline within digital forensics. It is field of process of data acquisition and analysis digital evidence from UAVs, by revealing sender and path of the drone [1].

2 Tools Used

A comprehensive drone forensic investigation requires a combination of commercial, proprietary suites and open-source analysis tools.

2.1 Category 1: Commercial & Proprietary Forensic Suites

These are the industry-standard tools used by law enforcement because they are reliable, but they are commercial so you need to pay to use them:

- **Cellebrite UFED & Physical Analyzer** [2]: These are primarily used for acquisition and Examination [2]. They extract data from the mobile devices (IOS/Android) connected to the drone controller [2].
- **Magnet AXIOM (Process & Examine)** [3]: This is another major proprietary tool used for Acquisition and Analysis [3]. It is unique because it can ingest both mobile data and drone-specific image dumps [3]. It can acquire and process evidence from multiple types of devices as: Android, IOS, Kindle Fire, Media devices, SIM cards [3].

2.2 Category 2: Open-Source & Freeware Analysis Tools

These tools are free or open-source and are often used for specific technical tasks that the big commercial tools might miss.

- **Autopsy** [4]: An Open-source platform used for general Examination and Analysis of hard drives and file systems [4]. It has a lot of modules like: Ingest modules, Hash Database Lookup Module and EXIF Parser module which can be specific and consist Multi-threaded and Priority [4].
- **DatCon** [5]: A highly specialized Freeware tool designed specifically for the "**Decryption of the .DAT files**" found on DJI drones [5]. This is critical because standard tools often cannot read these encrypted flight logs [5]. It has packages for exposing inconsistencies between the velocity and location values as IMUCalcs package, which utilize Flight Controllers as VeIN, VeIE, VeID, and creates a new data from `.DAT` [5].
- **ExifTool** [6]: An Open-source utility used for "**Reading meta information**" from photos and videos [6]. This helps investigators find GPS coordinates hidden inside images taken by the drone [6].

- **Binwalk** [7]: Used for "**Entropy measurement**" [7]. Binwalk can identify, and optionally extract, files and data that have been embedded inside of other files [7]. While its primary focus is firmware analysis, it supports a wide variety of file and data types [7].

While tools from Category 1 offer user-friendly interface, sometimes investigators need to use freeware as autopsy or DatCon to decrypt proprietary drone logs [5]. One case study that represent How tools can operate in drone forensics is DJI Mini 2 [8]. The necessity for a multi-tool approach in validation was realized in the study undertaken by Stanković et al. [8] using a DJI Mini 2. The authors simulated a criminal case where a drone captured surveillance footage [8]. They successfully showed that even though Cellebrite UFED was required to extract data from the connected iPhone 7 and Android devices, multiple tools were used to create the complete flight path and link the media evidence to the pilot [8].

3 Significant Challenges in Drone Forensics

The main challenges in drone forensics are coming from drone structure and standardization [9].

3.1 Lack of Standardization and Proprietary Formats

Unlike from simple computers drones are lack of standardization "UAV can host more than five different types of file systems which are undiagnosable by commercial tools [9]". Developing through new approaches for each drone model complicates the extraction process compared to other standards of technology [9]. The onboard drone's software, hardware and firmware are not standardized yet and they vary from one vendor to another [9].

3.2 Evidence Volatility and Anti-Forensic Capabilities

Another big issue is drone volatility and anti-forensic capabilities [9]. The drone's flight pass and remote connectivity are stored in volatile memory which means that memory will vanish as soon as battery runs out [1]. Some anti-forensic features also can be implemented into drone, as "remote wipeout feature" which means that memory can flow before it got to law enforcement [1]. The UAVs structure and components are small and versatile, and evidence can be scattered in various locations [9].

4 Solution for Challenges

The volatility of evidence, where flight data might be lost as soon as drone's battery runs out, and risk of remote wipeout, is addressed to be responded first [10].

4.1 Addressing Volatile Evidence and Remote Anti-Forensics

Solution for that, the drone must be isolated from all wireless networks to prevent remote wipeout and maintain power to keep versatile data [10]. After first respond, technical solution for system files involve bypassing potential restrictions to the operating system [10]. Investigators can establish the Telnet session with UAV to browse system directories [10].

4.2 Addressing Proprietary and Encrypted Data

The problem of manufacturer-specific proprietary file formats and encryption is barely overcome through reverse engineering and development of specialized open-source parsing tools [10]. Flight data extracted from the controller chip is normally encrypted [1]. To handle this, tools like **DatCon** were created to decrypt the `.DAT` files from DJI drones [5]. This methodology has

delivered more functionally advanced tools such as the **Drone Data Parser (DRDP)** [10], which is based on the principles of reverse engineering from DatCon and processes individual `DAT`, `TXT`, and default files [10] (See Figure 2 for the parsing process flow). With this process, it was known that the payload in the `.DAT` file structure itself is encrypted with an **XOR encryption algorithm** [10] (See Figure 3 for the `.DAT` file structure) [10]. By understanding and reverting this particular cryptographic method, forensic analysts can extract and decode essential data, bypassing its proprietary format into a readable CSV file for analysis [10].

5 Locations of Interest for Evidence Collection if Hacked Remotely

If a drone gets remotely hacked, evidence becomes scattered across different platforms [11].

- **UAV Firmware**: Tools like **Binwalk** [7] are required to check the firmware of the UAV for unauthorized code [11].
- **Onboard Flight Controller and Mobile Terminal Device**: These are important for the communication logs [11].
- **External Servers**: The drone may also be programmed to upload data to an external private or public server, and **cloud service data analysis** would be necessary for finding the attacker path and identity [11].

6 Conclusion

Drone forensics require a hybrid toolkit—commercial acquisition and open-source decryption—and strict first responder protocols to preserve volatile evidence [10]. Future data standardization and legal frameworks will determine the maturity of the discipline in retrieving cross-jurisdictional cloud evidence efficiently [11].

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